

Proposal Keyword Alignment Review

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Legend: - **[ADD: text]** = Insert this keyword/phrase from the RFA (shown in red) - **[REPLACE: old text → new text]** = Replace existing text with RFA-aligned language (shown in red) - **Blue terms** = Glossary terms (see Glossary at end of document)

Introduction

[ADD before first sentence: “In direct response to RFA DE-FOA-0003612, Challenge #6: Transforming Nuclear Restoration & Revitalization, this proposal addresses”]

DOE **Environmental Management (EM)** faces a persistent data-to-decision bottleneck. **[ADD after “bottleneck”: “DOE’s environmental cleanup mission faces an estimated \$540 billion liability over eight decades, with ~90 million gallons of highly radioactive tank waste requiring treatment and numerous assets requiring disposition that impedes site remediation and restoration crucial for revitalization of American energy, security, and innovation.”]** Data acquired from **environmental characterization**, monitoring, and site interpretation are often scattered among heterogeneous measurements, legacy reports, and site-specific documents, not ready for AI applications and informed decision-making. Consequently, high-value historical information, such as environmental characterization, detailed process sampling, **remediation**, monitoring, and site/facility operations datasets, remains difficult to discover, connect, and reuse in a timely way. We aim to fill this gap by curating **[REPLACE: “data sets” → “AI-Ready Datasets”]**, harmonizing multimodal data preparation and creating intelligent data retrieval and analytics via an AI-driven human-in-the loop **[REPLACE: “knowledge system” → “integrated knowledge management system”]** to accelerate EM decision-making and co-design waste management for the DOE complex with novel geospatial analysis and visualization enabled by AI. **[ADD: “This work directly supports Energy Dominance objectives and aligns with EM’s 2040 vision for significant liability reduction.”]**

This project will create a transferable and transformative AI-guided data and knowledge workflow that converts fragmented environmental and operational records into harmonized, verifiable information layers to support sensible **[REPLACE: “sensible decision-making” → “risk-informed decision-making”]** for DOE EM sites. This AI framework enhances

evidence retrieval performance, achieving higher Precision@k while remaining accountable cross-source data synthesis, and more effective **AI-enhanced** decision support for characterization, monitoring, remediation, and waste management workflows. We will use the well-established site operation data at ORNL and Y-12 in addition to background environmental data to establish the framework.^{1,2} **[ADD: “Drawing on DOE EM’s unparalleled 30+ years of operational data from unique nuclear processing facilities, the framework will also be designed for deployment at EM sites such as SRS and Hanford.”]** We will employ **exploratory data analysis (EDA)** to discover data trends,³ AI to predict potential anomalies,⁴ and an AI-guided workflow⁵ for multimodal data fusion, analysis, and scenario queries to support informed decision-making for DOE EM sites.

Project Objectives

The objective is to demonstrate that historical environmental measurements and legacy data and documents can be transformed into a harmonized and retrieval-ready information data structure that supports **[REPLACE: “informed decision workflows” → “risk-informed decision workflows and integrated performance and risk assessment”]** and provides a reliable foundation for broader cross-site and cross-media AI deployment. **[ADD: “This project is structured as a Seed Project under the Lighthouse Challenge framework, pressure-testing AI-driven workflows on well-characterized EM data before broader deployment.”]**

Phase I will: 1) establish a traceable, AI-guided workflow for pattern and anomaly discovery, organization, and harmonization of EM-relevant historical data and documents, 2) establish an integrated AI-assisted decision making framework that enables retrieval, synthesis, and use of these data assets to support waste management, storage, **[REPLACE: “and transport” → “transport, and waste disposition”]** and provide a foundation for broad deployment within DOE, and 3) create a bidirectional, **human-in-the-loop** interface that supports expert knowledge capture, user feedback, and workforce-facing delivery of AI-synthesized insights.

The Phase I project will validate the core workflow in a domain with a high technical readiness level (i.e., TRL>5) using air data. Phase II will extend the capabilities to water- and soil-related datasets and synchronize with the **[REPLACE: “R&D road map implementation” → “EM AI R&D Roadmap Implementation (Focus Area A)”]** in the DOE complex. The innovative AI platform proposed herein, offering data fusion, analytics, geospatial visualization, and decision-making, will position ORNL for leadership in implementing the **EM AI R&D Roadmap**. Utilizing the rich QA/QCed data curated at ORNL and Y-12 and the computational resources at ORNL (Frontier, **[ADD: “Genesis Mission supercomputing capabilities,”]** NERSC HPC) and UTK, we have a strong foundation to succeed in Phase I. **[ADD: “DOE/NSA National laboratory experts will**

partner with industry experts for accelerated simulation architecture in development of the AI models.”] Collaboration with UTK will provide training for the next-generation workforce in Tennessee’s booming nuclear industry, including fission and fusion, across TVA, Kairos, and Type One Energy.

[ADD new paragraph: “This effort is sponsored by the participating offices: the Office of Environmental Management (EM), Office of Chief Technology Officer (EM-3.2), and the Office of Science (SC), Advanced Scientific Computing Research (ASCR). The proposed work aligns with all three RFA Focus Areas: (A) EM AI R&D Roadmap Implementation, (B) Scale-Bridging AI Foundation Model development, and (C) Treatment Process Optimization at EM sites such as SRS and Hanford.”]

Proposed Research and Methods

The 9-month (M) effort will establish a proof-of-concept AI-guided data and knowledge workflow for the EM environmental data use case. We will perform three tasks.

Task 1. AI-Guided Data Curation and Harmonization (M1 - M4)

Task 1 directly supports the **[REPLACE: “Data Curation component of the AI R&D Roadmap” → “Data Curation component of the EM AI R&D Roadmap Implementation (Focus Area A)”]** and establishes the quality benchmarks and monitoring foundations needed for later deployment and improvement. **[ADD: “This task will enhance data accessibility across EM sites by converting historical data into standardized formats and evaluating multimodal data assets for AI applications, directly addressing the roadmap’s data-readiness goals.”]**

Task 1.1. AI-guided data discovery and organization. We will identify and organize historical data assets across three source tiers: (1) expert-curated foundational datasets, including NEPA **Environmental Impact Statements (EIS)** and associated EPA EIS database records, EM cleanup site operation data from ORCC, tabular environmental and operational datasets, and other authoritative historical records; (2) expert-selected literature and knowledge sources, including books, journal articles, and technical reports; and (3) MCP-enabled retrieval and crawl data used to supplement and update the knowledge base. For each source, AI agents will assist with query generation, document retrieval, source classification, duplicate detection, and relevance screening against predefined EM decision questions and priority variables. **[ADD: “Data assets will include operating & administrative procedures, characterization records spanning fractured systems, subsurface contamination, and liquid waste, as well as environmental liability and financial forecasts.”]** Because crawl data may be less reliable than expert-curated and operational sources, it will be assigned lower source-confidence scores and routed

through provenance review and expert validation before incorporation. An AI-guided workflow will be further developed to support source discovery, document screening, and initial relevance assessment based on our recent success in Automating Geospatial Web Dashboards Using Large Language Models and AI Agents for Decision Support.⁵ The primary outcome will be a prioritized inventory of candidate data assets, including source type, relevance score, confidence level, key variables, spatial/temporal coverage, operational relevance, and downstream use cases.

Task 1.2. Multimodal data harmonization and repository

development. Using the prioritized sources identified in Task 1.1, we will construct a harmonized, traceable, and extensible multimodal repository that links structured environmental and operational datasets with unstructured historical and technical records. **[ADD: “This repository will serve as the data foundation for a Scale-Bridging AI Foundation Model (Focus Area B) that predicts scale-dependent behavior across lab, pilot, and full-scale systems.”]** The initial air-domain repository will build on an existing ORR meteorological dataset containing structured measurements of temperature, pressure, humidity, wind, precipitation, and solar radiation. We will extend this site-based air dataset into an **[REPLACE: “AI-ready multimodal repository” → “AI-Ready Dataset repository”]** by linking structured monitoring data with document-derived context from EIS reports, agency records, technical reports, literature, and MCP-retrieved updates. Structured datasets will be harmonized by variable, unit, location, timestamp, and reporting period, while unstructured documents will be parsed into retrieval-ready segments and annotated with metadata, including source, document type, site/location, date, environmental medium, key variables, and decision context. A common harmonization schema will align records across variables, locations, time windows, document types, and provenance fields. Shared identifiers will connect air-domain measurements with relevant document segments, regulatory records, and technical knowledge. After completing the air-domain repository, we will assess candidate soil and water datasets based on data quality, quantity, metadata completeness, spatial-temporal coverage, and relevance to EM decision questions. Selected soil or water datasets will then be incorporated into the same Task 1 harmonization workflow to support cross-domain validation in Task 2.

Task 1.3. Data quality benchmarking, cataloging, and readiness

assessment. To support **[REPLACE: “AI-ready data preparation” → “AI-Ready Dataset preparation”]**, we will implement a lightweight quality engine for data cleaning, validation, deduplication, source-confidence assessment, and schema conformance during ingestion. Quality checks will assess completeness, spatial consistency, temporal coherence, provenance confidence, unit consistency, duplication, and cross-source linkage quality. Machine-readable metadata and cataloging will support discoverability, lineage tracing, retrieval readiness, and future incorporation of additional data from users and partners through common metadata and schema standards. **[ADD: “The quality engine will also support CARS analyses and best contract mechanisms by providing verifiable data lineage and quality metrics.”]** For the air-domain foundation, the prior ORR meteorological dataset already provides a quality-controlled starting point.

This project will build on those quality-control procedures and extend them into a broader AI-readiness framework that evaluates both structured datasets and document-derived knowledge sources. For soil and water datasets, readiness assessment will be used as a selection and screening step to determine whether the data are sufficient for harmonization, retrieval, analytics, and cross-domain testing. The overall outcome of Task 1 will be a harmonized, traceable, and knowledge-ready multimodal data repository, beginning with an AI-enhanced air-domain foundation and extending to selected soil and/or water datasets as data quality and availability allow. This repository, together with metadata, quality benchmarks, and readiness indicators, will provide the data-curation foundation for downstream retrieval, synthesis, analytics, deployment, and continuous-improvement workflows under the [EM AI R&D Roadmap](#).

Task 2. AI-Enhanced EM Analytics (M1 - M6)

Task 2 will build an analytics workflow for transforming the **[REPLACE: “AI-ready repository” → “AI-Ready Dataset repository”]** developed in Task 1 into **[REPLACE: “adaptive monitoring and early-warning support” → “adaptive & real-time monitoring and early warning systems for contamination and critical infrastructure failures”]** for EM decision-making.

Task 2.1 Air-domain adaptive monitoring and early-warning analytics.

Using the AI-ready repository developed in Task 1, we will develop an air-domain analytics workflow that builds on our prior EDA and machine-learning analyses of ORR air-domain data. This analytics workflow will include EDA to characterize temporal trends, spatial patterns, correlations among key air-related variables, and clustering patterns associated with extreme weather conditions. It will also include machine-learning models for predictive analytics, including forecasting selected air-quality indicators, identifying high-risk periods or locations, and evaluating variable importance. The workflow will support **[REPLACE: “adaptive and near-real-time monitoring concepts for early warning of contamination events and critical infrastructure or site-operation failures” → “adaptive & real-time monitoring for early warning systems for contamination and critical infrastructure failures, supporting integrated performance and risk assessment and risk-informed decision analyses”]**. **[ADD: “Analytics outputs will also support optimization of waste treatment processes, remedial systems, waste disposition, and site operations, directly contributing to Treatment Process Optimization (Focus Area C) at EM sites such as SRS and Hanford.”]**

Task 2.2 Cross-domain analytics validation using water/soil data. To evaluate whether the air-domain analytics workflow can be rapidly adapted to other EM-relevant environmental media, we will apply the workflow to representative water or soil datasets selected in Task 1. Domain-specific variables, such as contaminant concentration, sampling depth, environmental medium, sampling method, location, and reporting period, will be mapped to the shared schema and analyzed using the same

evaluation framework. We will compare results across air and water/soil to identify reusable workflow components, domain-specific adjustments, and modifications needed for Phase II scale-up. **[ADD: “Validation will also test the framework’s ability to support search for critical isotopes and characterization of fractured systems, subsurface contamination, and liquid waste across different environmental media.”]** This task will test whether the common schema, analytics-oriented feature construction, EDA procedures, and predictive modeling framework can be extended beyond the initial air-domain use case. Validation will assess whether the workflow can align records across locations and time periods, identify patterns and trends, support initial predictive analysis, and generate traceable outputs for EM monitoring and decision support in additional environmental media. **[ADD: “Cross-domain results will inform environmental liability and financial forecasts by providing validated predictive capabilities across multiple environmental media.”]** The outcome will be a validation summary documenting reusable components, domain-specific requirements, model-validation results, and recommendations for rapid deployment across additional EM sites and data domains in Phase II.

The overall outcome of Task 2 will be an **AI-enhanced** environmental monitoring and analytics workflow that transforms the AI-ready repository from Task 1 into interpretable and traceable support for EM decision-making. The workflow will support EM cleanup site operations, including environmental monitoring, remediation, nuclear fuel and waste management, **[REPLACE: “waste disposal, storage, transport” → “waste disposal, storage, transport, waste disposition”]**, and related mission activities. Task 2 results will provide the validated analytics foundation for the Task 3 human-in-the-loop decision-making framework.